

Multiple Choice (10 marks)

1. Which of the following will increase in vitamin D deficiency?
 - (a) The plasma concentration of vitamin B 12
 - (b) The plasma concentration of prothrombin
 - (c) The plasma concentration of preprothrombin
 - (d) The plasma activity of acid phosphatase
 - (e) The plasma activity of lipase
2. Which orbital wall is most likely to collapse in a 'blow out' fracture?
 - (a) The medial wall
 - (b) The ethmoidal wall
 - (c) The orbital rim
 - (d) The floor
 - (e) The roof
3. Blood flows from the right ventricle of the heart into which of the following structures?
 - (a) Coronary arteries
 - (b) Inferior vena cava
 - (c) Pulmonary capillaries
 - (d) Right atrium
 - (e) Left atrium
4. The initial length of telomere stand for members of a species is
 - (a) Determined by the individual's lifestyle habits
 - (b) Dependent on the individual's stress levels
 - (c) Randomly determined at birth
 - (d) Dependent on the individual's gender
 - (e) Dependent on the individual's diet and exercise
5. Men are more likely than women to die from
 - (a) Breast Cancer
 - (b) Influenza
 - (c) Stroke
 - (d) Suicide
 - (e) Asthma

True / False (10 marks)

Answer True or False

6. True or False: Regular aerobic exercise can significantly reduce the reaction time difference between younger and older adults.
7. True or False: The karyotype 47,XXX is associated with the mildest effects on human development among common trisomies.
8. True or False: Newborn infants in countries with limited access to safe water should be exclusively breastfed for the first six months of life.
9. True or False: Alcohol dehydrogenase is a zinc-dependent enzyme crucial for metabolizing ethanol in the liver.
10. True or False: Middle adapters are typically the most influential segment in driving community change efforts.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of vitamin E deficiency in the development of enamel defects and its correlation with an increased risk of dental caries. Analyze the underlying mechanisms involved.
12. Discuss the genetic implications of HLA haplotypes in familial relationships, particularly focusing on why sisters are likely to share a common HLA haplotype compared to other relatives.

End of Paper